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Subject Name: Database Management System

Week 6

Module 6 : SQL:

Faculty Name : UTTAM WAGHMODE

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Lecture 16 –SQL: SET OPERATIONS

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Lecture No: 16: SQL:SET OPERATIONS



SET OPERATIONS (UNION, INTERSECTION, MINUS/ DIFFERENCE)

The following rules must hold – R and S are two relations.

- R and S must have the same quantity of attributes.
- Attribute domains must be compatible.
- Duplicate tuples get automatically eliminated.

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UNION

It is used to combine the results of two or more SELECT statements.

Depositor (cust_name, account_no)

Borrower (cust_name, loan_no)

Loan (loan_no, branch, amount)

Depositor

Cust_name	Account_No
A	10
B	20
C	30
F	40

Borrower

Cust_name	Loan_No
D	101
B	102
C	103
E	104

Loan

Loan_No	Branch	Amount
101	Hyderabad	5000
102	Chennai	4000
103	Mumbai	7000
104	Hyderabad	2000

Activate Windows
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Query : Find names of the customers who have an account or loan or both.

```
SELECT Cust_name FROM Depositor
UNION
SELECT Cust_name FROM Borrower;
```

- OUTPUT WILL BE **A,B,C,F,D,E**

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INTERSECT

It is used to select common rows (tuples) from two tables (relations).

Query : Find names of the customers who have both account and loan.

```
SELECT Cust_name FROM Depositor
INTERSECT
SELECT Cust_name FROM Borrower;
```

- **OUTPUT WILL BE B,C**

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MINUS

MINUS (-):

Lets say we have two relations R1 and R2 and we want to select all those tuples(rows) that are present in Relation R1 but **not** present in Relation R2, this can be done using Set difference R1 – R2.

Query : Find names of the customers who have loan but not account.

```
SELECT Cust_name FROM Borrower
MINUS
SELECT Cust_name FROM Depositor;
```

- **OUTPUT WILL BE D,E**

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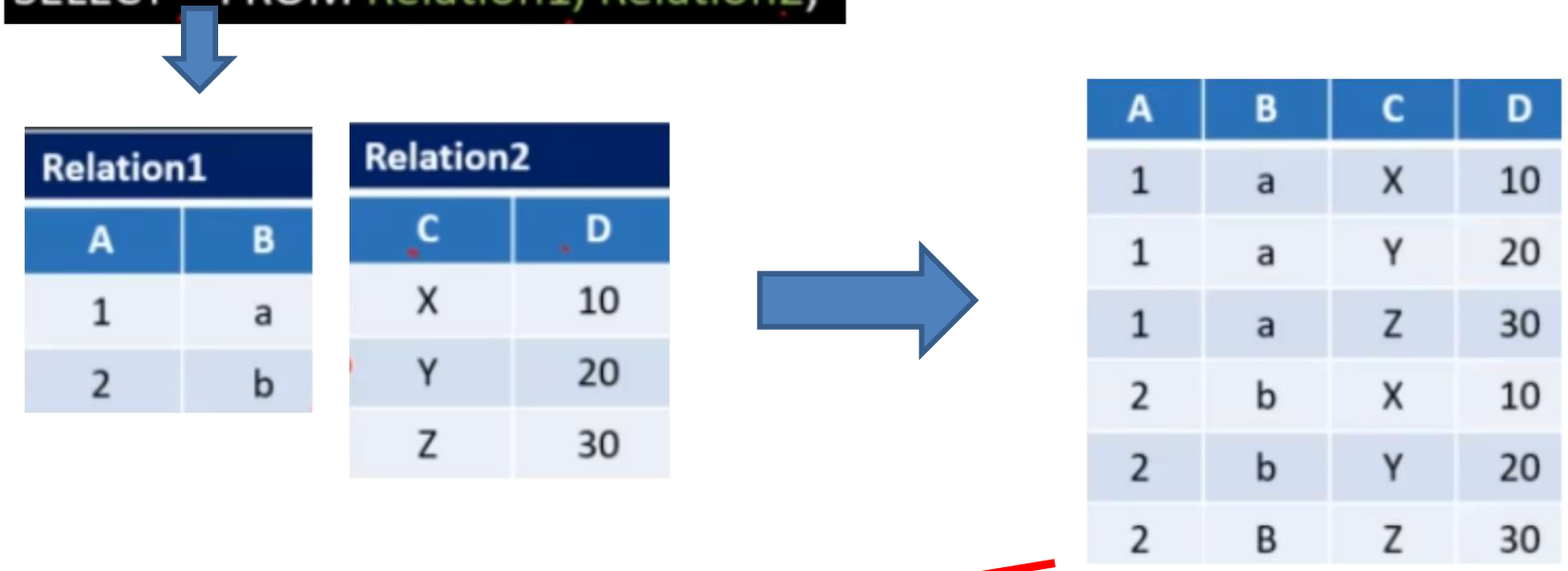
Lecture No: 17: SQL:JOIN OPERATIONS



JOINS:

Definition : A SQL Join statement is used to combine data or rows from two or more tables based on a common field between them.

```
SELECT * FROM Relation1, Relation2;
```



Relation1

A	B
1	a
2	b

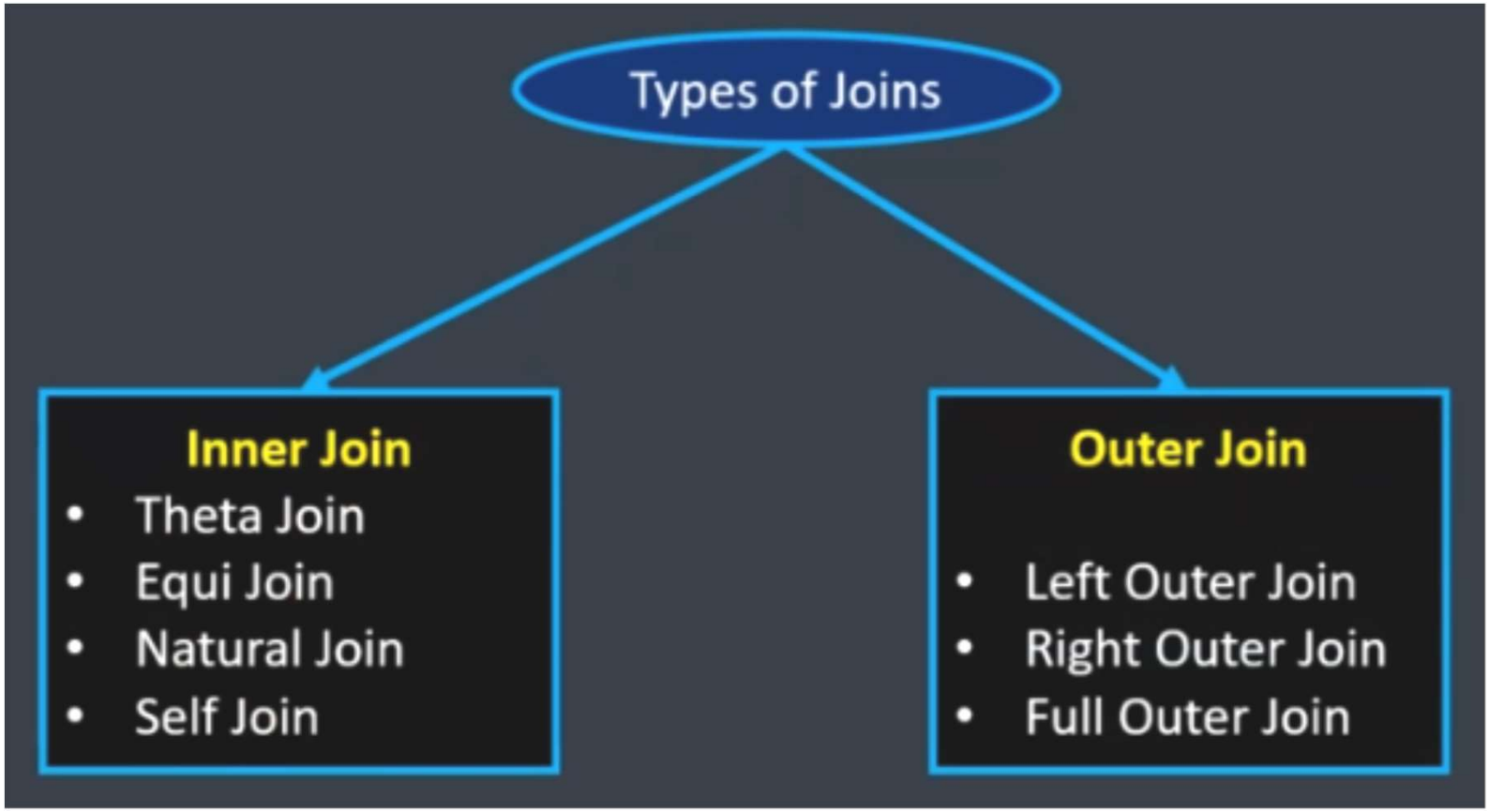
Relation2

C	D
X	10
Y	20
Z	30

A	B	C	D
1	a	X	10
1	a	Y	20
1	a	Z	30
2	b	X	10
2	b	Y	20
2	B	Z	30

CARTISIAN PRODUCT

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EQUI JOIN:

- **EQUI JOIN is done when the theta join uses only equivalence condition.**
- **Query 1: display the roll no of students who taken the course ‘DS’**

Student		
Rollno	cid	Marks
1	a	80
2	b	30
3	a	70
4	c	60

Course	
cid	cname
a	DS
b	DBMS
c	OS
d	CD



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Without C.cname = 'DS', the query would return all students with all courses.Adding this condition limits the results to only those students whose cid corresponds to the DS course.

- **Solution:**

```
SELECT S.Rollno, C.cname
FROM Student S, Course C
WHERE S.cid=C.cid AND C.cname= 'DS';
```

Rollno	cname
1	DS
3	DS



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NATURAL JOIN

- Natural join does not utilizing any of the comparison operator
- In natural join attributes should have same name and domain
- There should be at least one common attribute between two relations
- It performs selection forming equality on those attributes which appear in both the relations and eliminates duplicate attributes
- Query : display rollno, coursename of students

Student

Rollno	cid	Marks
1	a	80
2	b	30
3	a	70
4	c	60

Course

cid	cname
a	DS
b	DBMS
c	OS
d	CD



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- **SOLUTION :**

SELECT S.Rollno, C.cname

FROM Student S **NATURAL JOIN** Course C;

Rollno	cname
1	DS
2	DBMS
3	DS
4	OS



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- **Display rollno, course name of students who have taken the course 'DS'.**

Student		
Rollno	cid	Marks
1	a	80
2	b	30
3	a	70
4	c	60

Course	
cid	cname
a	DS
b	DBMS
c	OS
d	CD

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- **Solution:**

SELECT S.Rollno, C.cname

FROM Student S **NATURAL JOIN** Course C

WHERE C.cname='DS'

Rollno	cname
1	DS
3	DS

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SELF JOIN



- **Self join is join in which the table is join with itself**
- **Its also called unary relationships, specially when the table has FOREIGN KEY which references its own PRIMARY KEY.**

PRIMARY KEY

FOREIGN KEY

Employee			
empno	ename	mgr	Salary
1	Krishna		5000
2	Hari	1	2000
3	Ramu	2	4500
4	Karim	1	8000



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- **QUERY: LIST EACH EMPLOYEE MANAGER.**

Using FROM

Employee e1, employee e2: 4 X 4 OUTPUT OF CROSS PRODUCT for 1 row is shown in example

Employee			
empno	ename	mgr	Salary
1	Krishna		5000
2	Hari	1	2000
3	Ramu	2	4500
4	Karim	1	8000

Employee			
empno	ename	mgr	Salary
1	Krishna		5000
2	Hari	1	2000
3	Ramu	2	4500
4	Karim	1	8000

e1 e2

Employee1				Employee2			
empno	ename	mgr	Salary	empno	ename	mgr	Salary
2	Hari	1	2000	1	Krishna		5000
2	Hari	1	2000	2	Hari	1	2000
2	Hari	1	2000	3	Ramu	2	4500
2	Hari	1	2000	4	Karim	1	8000

If matching is there then only print the names e1.ename and e2.ename

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- **SOLUTION:**
- **SELECT e1.ename employee, e2.ename manager**
- **FROM Employee e1, employee e2**
- **WHERE e1.mgr=e2.empno;**
- **OUTPUT:**

employee	manager
Hari	Krishna
Ramu	Hari
Karim	Krishna



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Lecture No: 18: SQL: OUTER JOIN OPERATIONS



OUTER JOIN

- Outer join returns all rows from both participating tables which satisfy the join condition along with rows which do not satisfy the join condition.
- Types of outer join:

1. Left outer join
2. Right outer join
3. Full outer join

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The SQL LEFT JOIN (specified with the keywords LEFT JOIN and ON) joins two tables and fetches all matching rows of two tables for which the SQL-expression is true, plus rows from the FIRST table that do not match any row in the SECOND table.

Syntax :

```
SELECT *
FROM table1 LEFT JOIN table2
ON table1.column_name = table2.column_name;
```

Student

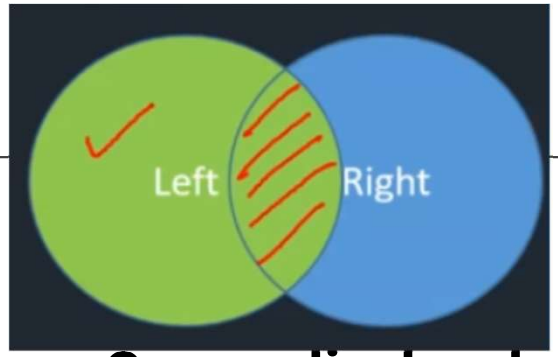
Rollno	Name	cid
1	Ravi	a
2	Krishna	b
3	John	a
4	Karim	c
5	Teja	
6	Srinu	

Course

cid	cname
a	DS
b	DBMS
c	OS
d	CD
e	TOC



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- **Query: display details of all students who taken course.**
-

Student

Rollno	Name	cid
1	Ravi	a
2	Krishna	b
3	John	a
4	Karim	c
5	Teja	
6	Srinu	

Course

cid	cname
a	DS
b	DBMS
c	OS
d	CD
e	TOC

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```
SELECT *
FROM Student LEFT JOIN Course
ON Student.cid=Course.cid;
```

Student

Rollno	Name	cid
1	Ravi	a
2	Krishna	b
3	John	a
4	Karim	c
5	Teja	
6	Srinu	

Course

cid	cname
a	DS
b	DBMS
a	DS
c	OS

NULL values are present

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RIGHT OUTER JOIN

The SQL RIGHT JOIN (specified with the keywords RIGHT JOIN and ON) joins two tables and fetches all matching rows of two tables for which the SQL-expression is true, plus rows from the SECOND table that do not match any row in the FIRST table.

Syntax :

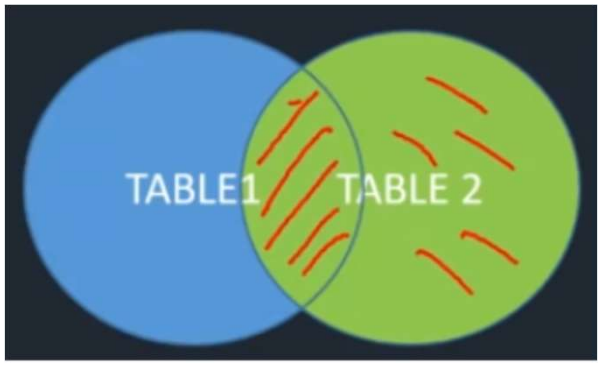
```
SELECT *
FROM table1 RIGHT JOIN table2
ON table1.column_name = table2.column_name;
```

Student

Rollno	Name	cid
1	Ravi	a
2	Krishna	b
3	John	a
4	Karim	c
5	Teja	
6	Srinu	

Course

cid	cname
a	DS
b	DBMS
c	OS
d	CD
e	TOC



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- **Query: display all courses details in output with each students what course they have taken.**

```
SELECT *
FROM Student RIGHT JOIN Course
ON Student.cid=Course.cid;
```

NULL values are present

Rollno	Name	cid	cid	cname
1	Ravi	a	a	DS
2	Krishna	b	b	DBMS
3	John	a	a	DS
4	Karim	c	c	OS
			d	CD
			e	TOC

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FULL OUTER JOIN

In SQL the FULL OUTER JOIN combines the results of both LEFT and RIGHT outer joins and returns all (matched or unmatched) rows from the tables on both sides of the join clause.

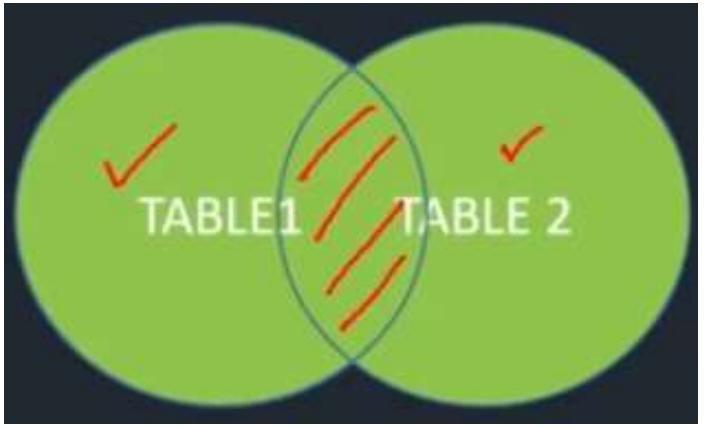
```
SELECT *
FROM table1 FULL OUTER JOIN table2
ON table1.column_name = table2.column_name;
```

Student

Rollno	Name	cid
1	Ravi	a
2	Krishna	b
3	John	a
4	Karim	c
5	Teja	
6	Srinu	

Course

cid	cname
a	DS
b	DBMS
c	OS
d	CD
e	TOC



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```
SELECT *
FROM Student FULL OUTER JOIN Course
ON Student.cid=Course.cid;
```

OUTPUT:

Student			Course	
Rollno	Name	cid	cid	cname
1	Ravi	a	a	DS
2	Krishna	b	b	DBMS
3	John	a	a	DS
4	Karim	c	c	OS
5	Teja	NULL	NULL	NULL
6	Srinu	NULL	NULL	NULL
NULL	NULL	NULL	d	CD
NULL	NULL	NULL	e	TOC

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Thank You



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